

Spark Homes Field Estimator - One Page Writeup

Devarsh Vora | Mobile-first field estimator for acquisition walkthroughs

1. Most interesting design or UX decision

I designed the app around the order of a real property walkthrough instead of copying a desktop spreadsheet into a phone screen. The main repair total, progress, current rooms, and add-room action stay close to the agent because those are the pieces that matter while moving through a house. Rooms behave like field objects, so an agent can add Bathroom 2, Bedroom 3, or a Living/Common Area as they see the layout, and each room keeps its own repair groups, quantities, notes, and photos. I also separated Living/Common Areas from Interior/General because hallways and living rooms are easy places to hide flooring, paint, door, and lighting scope if they are treated as a catch-all section. The UI uses large touch targets, short labels, collapsible groups, and simple language so the app can be used quickly with one hand while standing in a property.

2. What is broken or fragile in the build

The biggest fragile area is local browser storage. The app is static and offline-ready, but photos are stored locally, which can become heavy if an agent captures many full-resolution images. A production version should compress photos before saving and should eventually sync completed projects to a shared team system after the phone is back online. Serial number handling is also limited. The app makes a basic attempt to help with serial evidence, but it does not perform true OCR on HVAC, water heater, or appliance plates. Export is designed to keep working offline, but the best spreadsheet formatting depends on the browser having the optional spreadsheet library available. I kept these tradeoffs because the contest version needed to stay simple, installable, and reliable without a backend or setup step.

3. Creative addition and why I chose it

I added a Tools section with three practical features: Offer Worksheet, Offer Risk Check, and Walkthrough Guide. Offer Worksheet connects the repair estimate to acquisition math. The agent can enter ARV, purchase price, closing or holding costs, and immediately see whether the numbers still leave room for profit. Offer Risk Check is meant to slow the agent down only when it matters. It looks at high-cost selected repairs, unfinished groups, missing photo evidence, and the size of the repair estimate so the agent can decide if something needs another look before leaving. Walkthrough Guide is different from the offer tools because it is about the physical walk itself. It points the agent toward the next unfinished room or group, estimates what is left, and keeps the checklist from becoming a long scroll with no direction. I chose these three tools because a repair estimator should not only total line items. It should help the agent leave the house with a cleaner scope, enough evidence, and a better decision on whether the offer is worth pursuing.

4. What I would ship next with two more days

With two more days, I would first improve the photo workflow. I would add automatic image compression, a dedicated serial plate capture prompt for systems and appliances, and real OCR for HVAC, water heaters, and appliance labels. I would also add a share-ready scope narrative that turns the selected repairs, room notes, and photos into a short summary for the acquisitions or construction team. That would reduce the gap between a field estimate and a contractor handoff. The next improvement would be a project comparison view for acquisition leads, showing repair cost per square foot, high-risk categories, and estimate differences across recent properties. This would make it easier to notice when one estimate looks unusually high or low compared with similar deals.

Role of AI tools in development

I used AI tools as a development assistant for iteration, code review, and edge-case checking, especially around mobile layout, offline behavior, export, and wording. The product decisions were driven by the brief and by how an agent would use the app during a walkthrough: fast room setup, visible totals, clear progress, photo evidence, and a clean export. I reviewed the final behavior and adjusted the interface so it would feel like a field tool rather than a presentation demo. I used the reference app to understand scope only, then rebuilt the experience around the submitted workflow and design.